

Sampling with Schrödinger Bridge

An Adjoint-Matching Perspective

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 Meta

 Georgia
Tech

Outline

- Sampling problem setup
- Diffusion sampler
- Adjoint-based diffusion sampler
- Adjoint Schrodinger Bridge Sampler (*NeurIPS 2025 Oral*)

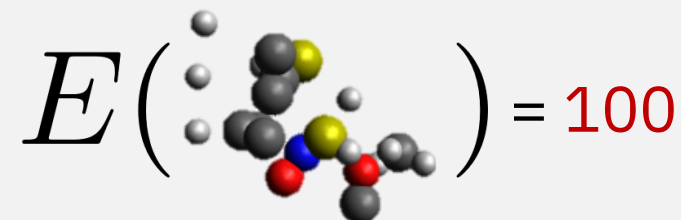
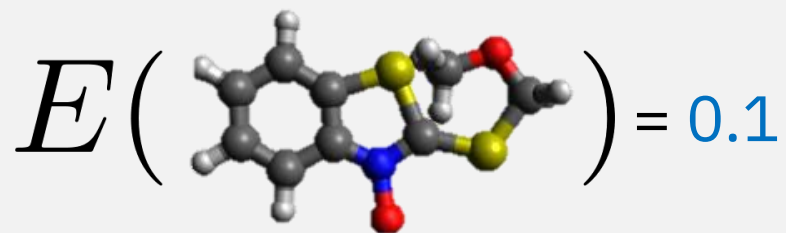
Sampling Problem

Boltzmann Distribution

$$\rho_{\text{target}}(x) := \frac{1}{Z} \exp(-E(x)), \quad Z := \int_{\mathcal{X}} \exp(-E(x)) dx$$

unnormalized energy function
(differentiable)

normalization constant
(usually intractable)



low energy → stable structure → likely to appear → high probability
high energy → unstable structure → unlikely to appear → low probability

Sampling Problem

Boltzmann Distribution

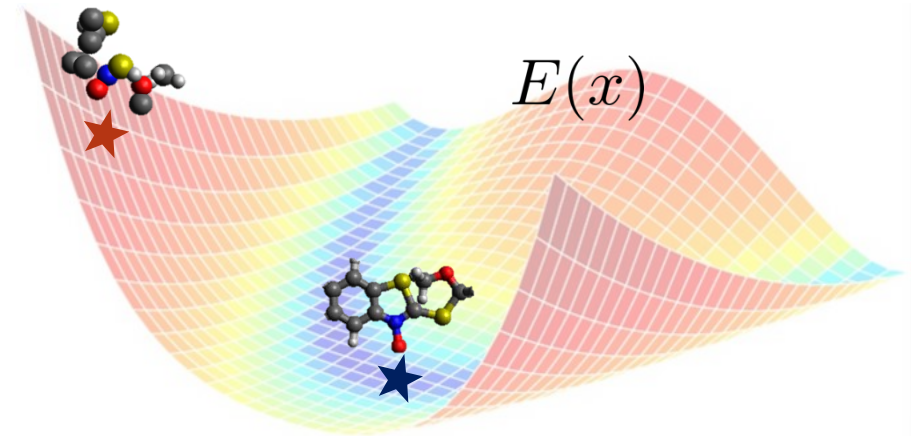
$$\rho_{\text{target}}(x) := \frac{1}{Z} \exp(-E(x)), \quad Z := \int_{\mathcal{X}} \exp(-E(x)) dx$$

unnormalized energy function
(differentiable)

normalization constant
(usually intractable)

Problem Statement

Given $E(x)$, generate $X \sim \rho_{\text{target}}$.



Sampling Method

Given $E(x)$, generate $X \sim \rho_{\text{target}}$, where $\rho_{\text{target}}(x) := \frac{1}{Z} \exp(-E(x))$

MCMC methods

Metropolis-Hastings
Annealed Importance Sampling
Hamiltonian Monte Carlo

...

Metropolis-Adjusted Langevin Algorithm

$$dX_t = \underbrace{-\nabla E(X_t)dt}_{\text{exploitation}} + \underbrace{\sqrt{2}dW_t}_{\text{exploration}}$$

Learning-based methods

Diffusion samplers

$$dX_t = [\underbrace{f_t(X_t)}_{\text{prescribed base drift}} + \underbrace{u_t^\theta(X_t)}_{\text{learned control drift}}]dt + \underbrace{\sigma_t}_{\text{prescribed noise schedule}}dW_t,$$

$$X_0 \sim \rho_{\text{source}}$$

❌ slow mixing rate

❌ expansive energy evaluation

Diffusion Sampler

$$X_0 \sim \rho_{\text{source}}$$

Find $u_t^\theta(\cdot)$ such that $X_1 \sim \rho_{\text{target}}$, where $dX_t = [f_t(X_t) + u_t^\theta(X_t)]dt + \sigma_t dW_t$,

$$\rho_{\text{target}}(x) := \frac{1}{Z} \exp(-E(x))$$

given

prescribed

Published in Transactions on Machine Learning Research (02/2024)

Published

Published as a conference paper at ICLR 2023

Published as a conference paper at ICI

An op

on dif

DENOISING DIFFUSION SAMPLERS

FLO

BOOT

STRAP

TRANSPORT MEETS VARIATIONAL

SEQUENTIAL CONTROLLED LANGEVIN DIFFUSIONS

CONTROLLED MONTE CARLO

Particle Denoising Diffusion Sampler

Iterated Denoising Energy Matching for Sampling

Diffusion Sampler

Find $u_t^\theta(\cdot)$ such that $X_1 \sim \rho_{\text{target}}$, where $X_0 \sim \rho_{\text{source}}$
$$dX_t = [f_t(X_t) + u_t^\theta(X_t)]dt + \sigma_t dW_t,$$
$$\rho_{\text{target}}(x) := \frac{1}{Z} \exp(-E(x))$$

given

prescribed

Matching methods

$$\arg \min_{\theta} \mathbb{E} [\|u_t^\theta(X_t) - \square\|^2]$$



scalable

Non-Matching methods

$$\nabla_{\theta} \mathcal{L}(X_1) \quad \frac{dp^u(X)}{dq(X)}$$



backpropagation through SDE,
higher-order derivatives

Diffusion Sampler

Find $u_t^\theta(\cdot)$ such that $X_1 \sim \rho_{\text{target}}$, where $X_0 \sim \rho_{\text{source}}$
$$dX_t = [f_t(X_t) + u_t^\theta(X_t)]dt + \sigma_t dW_t,$$
$$\rho_{\text{target}}(x) := \frac{1}{Z} \exp(-E(x))$$

given

prescribed

Matching methods

$$\arg \min_{\theta} \mathbb{E} [\|u_t^\theta(X_t) - \square\|^2]$$

score/flow/bridge matching

✗ requires $X_1 \sim \rho_{\text{target}}$

adjoint matching

✓ does NOT require $X_1 \sim \rho_{\text{target}}$

Adjoint Matching

Method for Stochastic Optimal Control (SOC)

$$\begin{aligned} & \min_u \mathbb{E} \left[\int_0^1 \frac{1}{2\sigma_t^2} \|u_t(X_t)\|^2 dt + g(X_1) \right] \\ & \text{s.t. } dX_t = [f_t(X_t) + u_t(X_t)]dt + \sigma_t dW_t, \quad X_0 \sim \rho_{\text{source}} \end{aligned}$$

↕

$$u^* = \arg \min_u \mathbb{E}_{p_{\bar{u}}} [\|u_t(X_t) - \Phi_t(\mathbf{X}) \cdot \nabla g(X_1)\|^2]$$

$\bar{u} = \text{stopgrad}(u)$

- ✓ Matching objective
- ✓ Does *not* require $X_1 \sim \rho_{\text{target}}$
- ✓ Detached model samples

Adjoint Matching

Method for Stochastic Optimal Control (SOC)

$$\begin{aligned} \min_u \mathbb{E} \left[\int_0^1 \frac{1}{2\sigma_t^2} \|u_t(X_t)\|^2 dt + g(X_1) \right] \\ \text{s.t. } dX_t = [f_t(X_t) + u_t(X_t)]dt + \sigma_t dW_t, \quad X_0 \sim \rho_{\text{source}} \end{aligned}$$



$$\begin{aligned} u^* = \arg \min_u \mathbb{E}_{p^{\bar{u}}} [\|u_t(X_t) - \Phi_t(\mathbf{X}) \cdot \nabla g(X_1)\|^2] \\ \bar{u} = \text{stopgrad}(u) \end{aligned}$$



$$p^*(X_1|X_0) \propto p^{\text{base}}(X_1|X_0) e^{-g(X_1)}$$

optimal distribution
induced by u^*

base
distribution

exponentially
tilted

$$dX_t = \underline{f_t(X_t)}dt + \underline{\sigma_t}dW_t, \quad X_0 \sim \underline{\rho_{\text{source}}}$$

Adjoint Matching

Application for sampling

$$\min_u \mathbb{E} \left[\int_0^1 \frac{1}{2\sigma_t^2} \|u_t(X_t)\|^2 dt + g(X_1) \right]$$

s.t. $dX_t = [f_t(X_t) + u_t(X_t)]dt + \sigma_t dW_t, \quad X_0 \sim \rho_{\text{source}}$



$$u^* = \arg \min_u \mathbb{E}_{p^{\bar{u}}} [\|u_t(X_t) - \Phi_t(\mathbf{X}) \cdot \nabla g(X_1)\|^2]$$

$\bar{u} = \text{stopgrad}(u)$



$$p^*(X_1|X_0) \propto p^{\text{base}}(X_1|X_0) e^{-g(X_1)}$$

Design g such that $\rho_{\text{target}}(X_1) = \int p^*(X_1|X_0) \rho_{\text{source}}(X_0) dX_0$

This is Schrödinger Bridge!

Connection to Schrödinger Bridge

SOC Optimal Solution

$$p^*(X_0, X_1) = p^{\text{base}}(X_0, X_1) e^{-g(X_1) + V_0(X_0)}, \quad V_0(x) = -\log \int p_{1|0}^{\text{base}}(y|x) e^{-g(y)} dy$$



SOC Optimal Solution (reparametrized)

$$p^*(X_0, X_1) = p^{\text{base}}(X_0, X_1) \frac{\varphi_1(X_1)}{\varphi_0(X_0)}, \quad \varphi_0(x) = \int p_{1|0}^{\text{base}}(y|x) \varphi_1(y) dy, \quad \varphi_1(\cdot) = e^{-g(\cdot)}$$

Design g such that $\rho_{\text{target}}(X_1) = \int p^*(X_0, X_1) dX_0$

$$\Leftrightarrow \rho_{\text{target}}(X_1) = \varphi_1(X_1) \int p^{\text{base}}(X_1|X_0) \frac{\rho_{\text{prior}}(X_0)}{\varphi_0(X_0)} dX_0$$

$$\Leftrightarrow \frac{\rho_{\text{target}}(X_1)}{\varphi_1(X_1)} = \int p^{\text{base}}(X_1|X_0) \frac{\rho_{\text{prior}}(X_0)}{\varphi_0(X_0)} dX_0$$

$$\Leftrightarrow \hat{\varphi}_1(X_1) = \int p^{\text{base}}(X_1|X_0) \hat{\varphi}_0(X_0) dX_0$$

Connection to Schrödinger Bridge

SOC Optimal Solution

$$p^*(X_0, X_1) = p^{\text{base}}(X_0, X_1)e^{-g(X_1)+V_0(X_0)}, \quad V_0(x) = -\log \int p_{1|0}^{\text{base}}(y|x)e^{-g(y)}dy$$

Design g such that $\rho_{\text{target}}(X_1) = \int p^*(X_0, X_1)dX_0$

Conditions to Sample Target Distr.

Suppose $\varphi_1(\cdot) = e^{-g(\cdot)}$ satisfy

$$\begin{aligned} \varphi_0(x) &= \int p_{1|0}^{\text{base}}(y|x)\varphi_1(y)dy, & \varphi_0(x)\hat{\varphi}_0(x) &= \rho_{\text{prior}}(x), \\ \hat{\varphi}_1(y) &= \int p_{1|0}^{\text{base}}(y|x)\hat{\varphi}_0(x)dx, & \varphi_1(y)\hat{\varphi}_1(y) &= \rho_{\text{target}}(y), \end{aligned}$$

SB Theory: There's a *unique* g such that these equations hold.

(up to additive constant)

Adjoint-based Diffusion Sampler

Variational Inference
(Diffusion models → Stochastic Optimal Control)



Scalable Computational Method
(SOC → adjoint matching)



Probabilistic Characteristics

Adjoint Schrödinger Bridge Sampler

Variational Inference
(Diffusion models → Stochastic Optimal Control)



Scalable Computational Method
(SOC → adjoint matching)



$$\Rightarrow p^*(X_1|X_0) \propto p^{\text{base}}(X_1|X_0)e^{-g(X_1)}$$

$$g(X_1) := \log \hat{\varphi}_1(X_1) + E(X_1)$$

$$\varphi_0(x) = \int p_{1|0}^{\text{base}}(y|x)\varphi_1(y)dy, \quad \varphi_0(x)\hat{\varphi}_0(x) = \rho_{\text{prior}}(x),$$

$$\hat{\varphi}_1(y) = \int p_{1|0}^{\text{base}}(y|x)\hat{\varphi}_0(x)dx, \quad \varphi_1(y)\hat{\varphi}_1(y) = \rho_{\text{target}}(y),$$

Adjoint Schrödinger Bridge Sampler

$$\begin{aligned} \min_u \mathbb{E} & \left[\int_0^1 \frac{1}{2\sigma_t^2} \|u_t(X_t)\|^2 dt + \log \hat{\varphi}_1(X_1) + E(X_1) \right] \\ \text{s.t. } dX_t &= u_t(X_t)dt + \sigma_t dW_t, \quad X_0 \sim \rho_{\text{source}} \end{aligned}$$



Scalable Computational Method
(SOC → adjoint matching)



$$\Rightarrow p^*(X_1|X_0) \propto p^{\text{base}}(X_1|X_0) e^{-g(X_1)}$$

$$g(X_1) := \log \hat{\varphi}_1(X_1) + E(X_1)$$

$$\varphi_0(x) = \int p_{1|0}^{\text{base}}(y|x) \varphi_1(y) dy, \quad \varphi_0(x) \hat{\varphi}_0(x) = \rho_{\text{prior}}(x),$$

$$\hat{\varphi}_1(y) = \int p_{1|0}^{\text{base}}(y|x) \hat{\varphi}_0(x) dx, \quad \varphi_1(y) \hat{\varphi}_1(y) = \rho_{\text{target}}(y),$$

Adjoint Schrödinger Bridge Sampler

$$\min_u \mathbb{E} \left[\int_0^1 \frac{1}{2\sigma_t^2} \|u_t(X_t)\|^2 dt + \text{intractable} \log \hat{\varphi}_1(X_1) + E(X_1) \right]$$

$$\text{s.t. } dX_t = u_t(X_t)dt + \sigma_t dW_t, \quad X_0 \sim \rho_{\text{source}}$$



$$u^* = \arg \min_u \mathbb{E}_{p^{\bar{u}}} [\|u_t(X_t) + \sigma_t^2 (\nabla E + \nabla \log \hat{\varphi}_1)(X_1)\|^2]$$

$$\nabla \log \hat{\varphi}_1 = \arg \min_h \mathbb{E}_{p^{\bar{u}}} [\|h(X_1) - \nabla_{x_1} \log p^{\text{base}}(X_1|X_0)\|^2]$$

$$\bar{u} = \text{stopgrad}(u)$$



$$p^*(X_1|X_0) \propto p^{\text{base}}(X_1|X_0) e^{-g(X_1)}$$

$$g(X_1) := \log \hat{\varphi}_1(X_1) + E(X_1)$$

$$\varphi_0(x) = \int p_{1|0}^{\text{base}}(y|x) \varphi_1(y) dy, \quad \varphi_0(x) \hat{\varphi}_0(x) = \rho_{\text{prior}}(x),$$

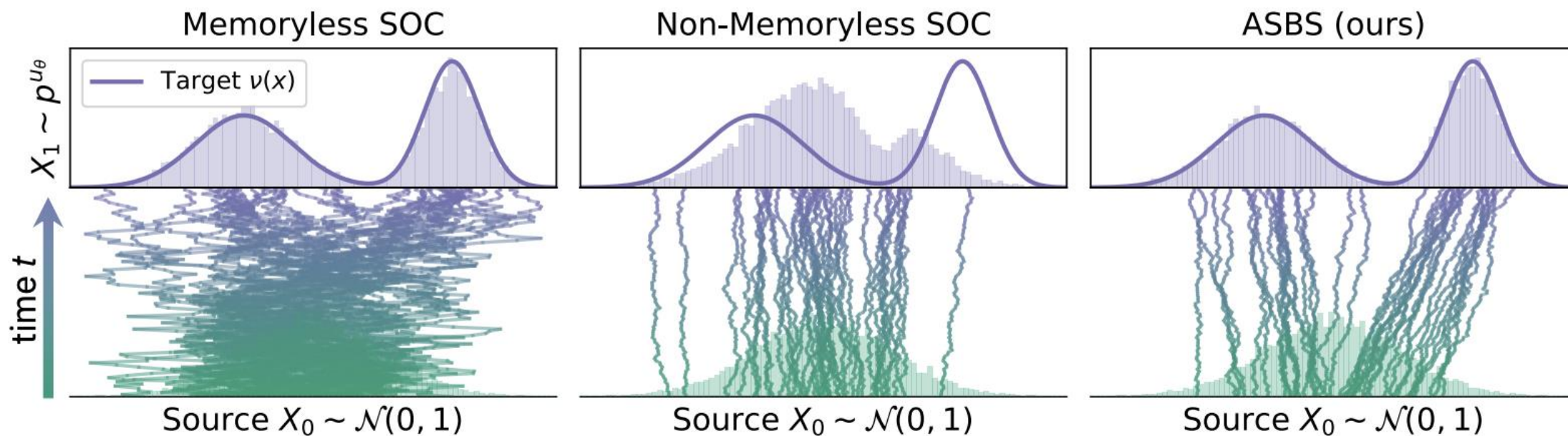
$$\hat{\varphi}_1(y) = \int p_{1|0}^{\text{base}}(y|x) \hat{\varphi}_0(x) dx, \quad \varphi_1(y) \hat{\varphi}_1(y) = \rho_{\text{target}}(y),$$

Adjoint Matching

Corrector Matching



Adjoint Schrödinger Bridge Sampler



Adjoint Schrödinger Bridge Sampler

	MW-5 ($d=5$)	DW-4 ($d=8$)		LJ-13 ($d=39$)			LJ-55 ($d=165$)		
Method	Sinkhorn $\downarrow$	$\mathcal{W}_2 \downarrow$	$E(\cdot)$ $\mathcal{W}_2 \downarrow$	$\mathcal{W}_2 \downarrow$	$E(\cdot)$ $\mathcal{W}_2 \downarrow$	$\mathcal{W}_2 \downarrow$	$E(\cdot)$ $\mathcal{W}_2 \downarrow$	$\mathcal{W}_2 \downarrow$	$E(\cdot)$ $\mathcal{W}_2 \downarrow$
PDDS (Phillips et al., 2024)	—	0.92 ± 0.08	0.58 ± 0.25	4.66 ± 0.87	56.01 ± 10.80	—	—	—	—
SCLD (Chen et al., 2025)	0.44 ± 0.06	1.30 ± 0.64	0.40 ± 0.19	2.93 ± 0.19	27.98 ± 1.26	—	—	—	—
PIS (Zhang and Chen, 2022)	0.65 ± 0.25	0.68 ± 0.28	0.65 ± 0.25	1.93 ± 0.07	18.02 ± 1.12	4.79 ± 0.45	228.70 ± 131.27	—	—
DDS (Vargas et al., 2023)	0.63 ± 0.24	0.92 ± 0.11	0.90 ± 0.37	1.99 ± 0.13	24.61 ± 8.99	4.60 ± 0.09	173.09 ± 18.01	—	—
LV-PIS (Richter and Berner, 2024)	—	1.04 ± 0.29	1.89 ± 0.89	—	—	—	—	—	—
iDEM (Akhound-Sadegh et al., 2024)	—	0.70 ± 0.06	0.55 ± 0.14	1.61 ± 0.01	30.78 ± 24.46	4.69 ± 1.52	93.53 ± 16.31	—	—
AS (Havens et al., 2025)	0.32 ± 0.06	0.62 ± 0.06	0.55 ± 0.12	1.67 ± 0.01	2.40 ± 1.25	4.50 ± 0.05	58.04 ± 20.98	—	—
ASBS (Ours)	0.15 ± 0.02	0.38 ± 0.05	0.19 ± 0.03	1.59 ± 0.00	1.28 ± 0.22	4.00 ± 0.03	27.69 ± 3.86	—	—

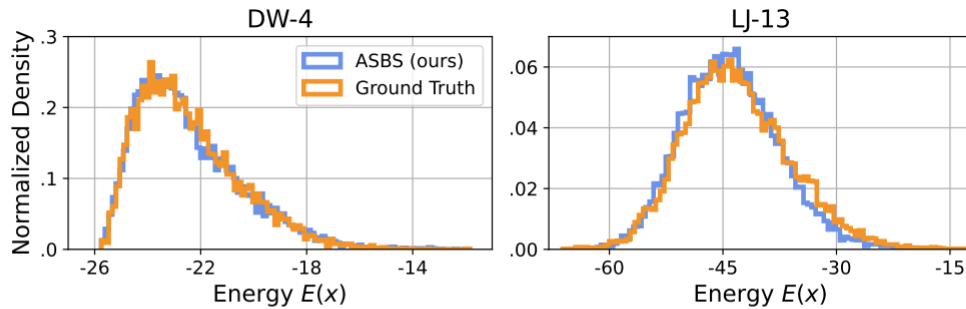


Figure 3 The energy histograms of DW-4 and LJ-13 from Table 2. ASBS generates samples whose energy profiles closely match those of the ground-truth samples.

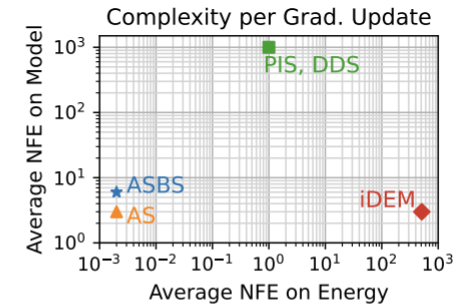


Figure 4 Complexity w.r.t. the number of function evaluation (NFE) on LJ-13 potential.

Adjoint Schrödinger Bridge Sampler

Large-Scale Amortized Conformer Generation

$$\rho_{\text{target}}(x|g) = e^{-\frac{1}{\tau} E(x|g)}, \quad \begin{array}{c} g \in \mathcal{G}, \\ \text{graph structure} \end{array} \quad \begin{array}{c} \tau \ll 1 \\ \text{conformer} \end{array}$$

Method	without relaxation				with relaxation			
	SPICE		GEOM-DRUGS		SPICE		GEOM-DRUGS	
	Coverage $\uparrow$	AMR $\downarrow$	Coverage $\uparrow$	AMR $\downarrow$	Coverage $\uparrow$	AMR $\downarrow$	Coverage $\uparrow$	AMR $\downarrow$
RDKit ETKDG (Riniker and Landrum, 2015)	56.94 $\pm$ 35.82	1.04 $\pm$ 0.52	50.81 $\pm$ 34.69	1.15 $\pm$ 0.61	70.21 $\pm$ 31.70	0.79 $\pm$ 0.44	62.55 $\pm$ 31.67	0.93 $\pm$ 0.53
AS (Havens et al., 2025)	56.75 $\pm$ 38.15	0.96 $\pm$ 0.26	36.23 $\pm$ 33.42	1.20 $\pm$ 0.43	82.41 $\pm$ 25.85	0.68 $\pm$ 0.28	64.26 $\pm$ 34.57	0.89 $\pm$ 0.45
ASBS w/ Gaussian prior (Ours)	73.04 $\pm$ 31.95	0.83 $\pm$ 0.24	50.23 $\pm$ 35.98	1.05 $\pm$ 0.43	88.26 $\pm$ 20.57	0.60 $\pm$ 0.24	72.32 $\pm$ 29.68	0.77 $\pm$ 0.35
ASBS w/ harmonic prior (Ours)	74.05 $\pm$ 31.61	0.82 $\pm$ 0.23	53.14 $\pm$ 35.69	1.03 $\pm$ 0.42	88.71 $\pm$ 18.63	0.59 $\pm$ 0.24	72.77 $\pm$ 29.94	0.78 $\pm$ 0.35

Adjoint Schrödinger Bridge Sampler

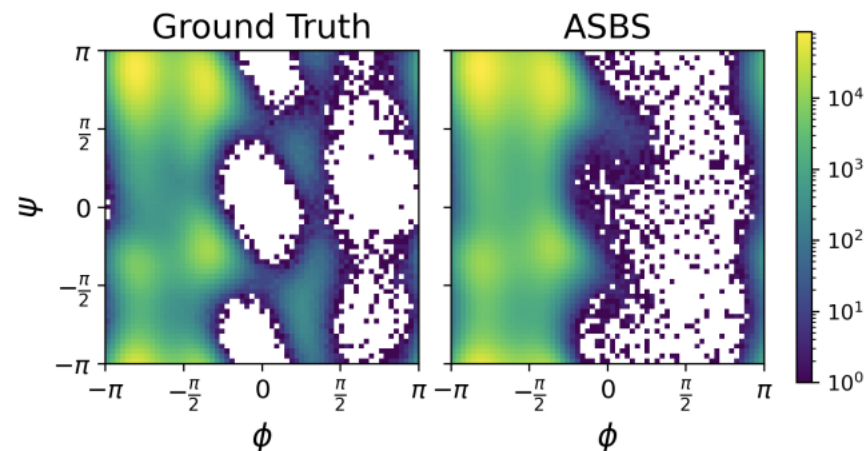
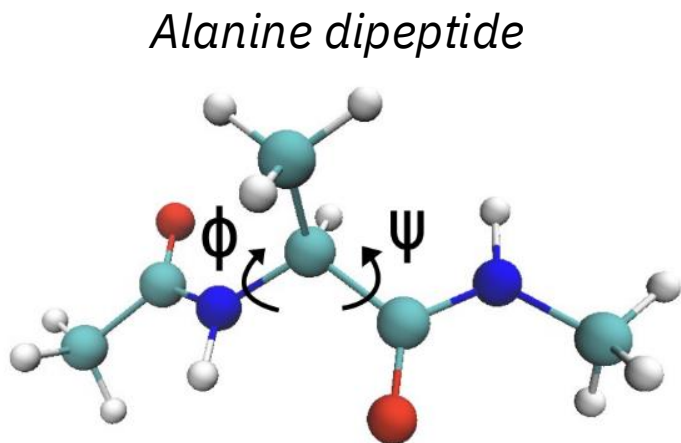
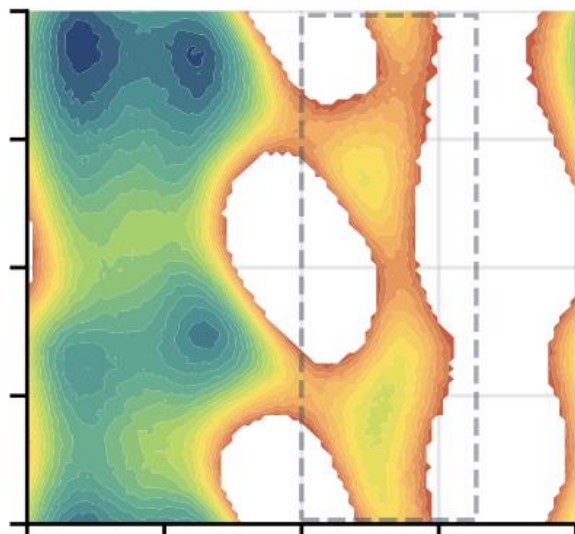
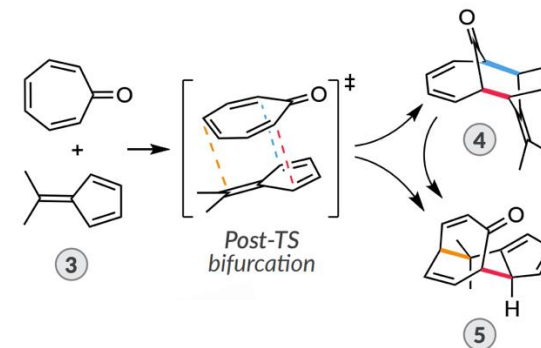
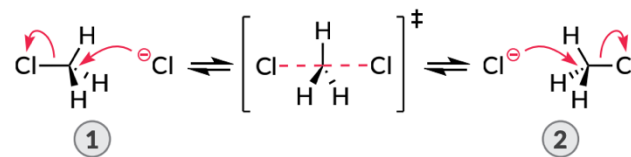


Table 3 Comparison between diffusion samplers on sampling the molecular Boltzmann distribution of the alanine dipeptide. We report the KL divergence D_{KL} for the 1D marginal across five torsion angles and the Wasserstein-2 $\mathcal{W}_2$ on jointly (ϕ, ψ) , known as Ramachandran plots (see Figure 5). Best results are highlighted.

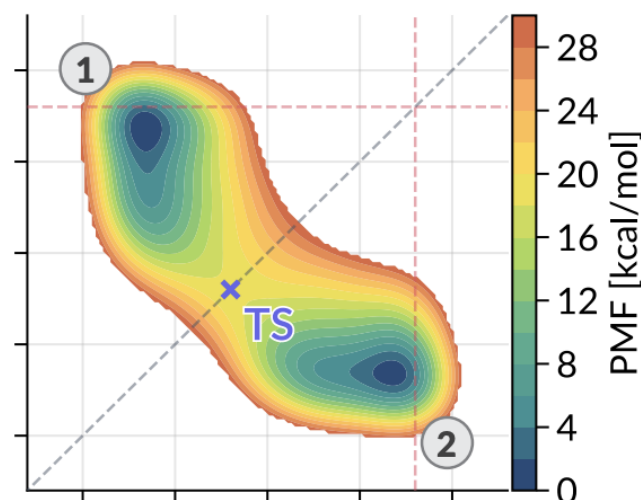
Method	D_{KL} on each torsion's marginal ↓					$\mathcal{W}_2$ on joint ↓
	ϕ	ψ	γ_1	γ_2	γ_3	(ϕ, ψ)
PIS (Zhang and Chen, 2022)	0.05 ± 0.03	0.38 ± 0.49	5.61 ± 1.24	4.49 ± 0.03	4.60 ± 0.03	1.27 ± 1.19
DDS (Vargas et al., 2023)	0.03 ± 0.01	0.16 ± 0.07	2.44 ± 0.96	0.03 ± 0.00	0.03 ± 0.00	0.68 ± 0.09
AS (Havens et al., 2025)	0.09 ± 0.09	0.04 ± 0.04	0.17 ± 0.17	0.56 ± 0.09	0.51 ± 0.06	0.65 ± 0.52
ASBS (Ours)	0.02 ± 0.00	0.01 ± 0.00	0.03 ± 0.01	0.02 ± 0.00	0.02 ± 0.00	0.25 ± 0.01

Adjoint Schrödinger Bridge Sampler

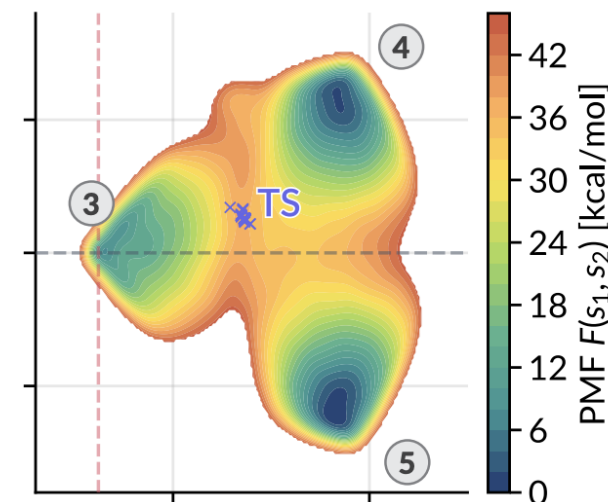
Adding chemical-based exploration



Alanine dipeptide



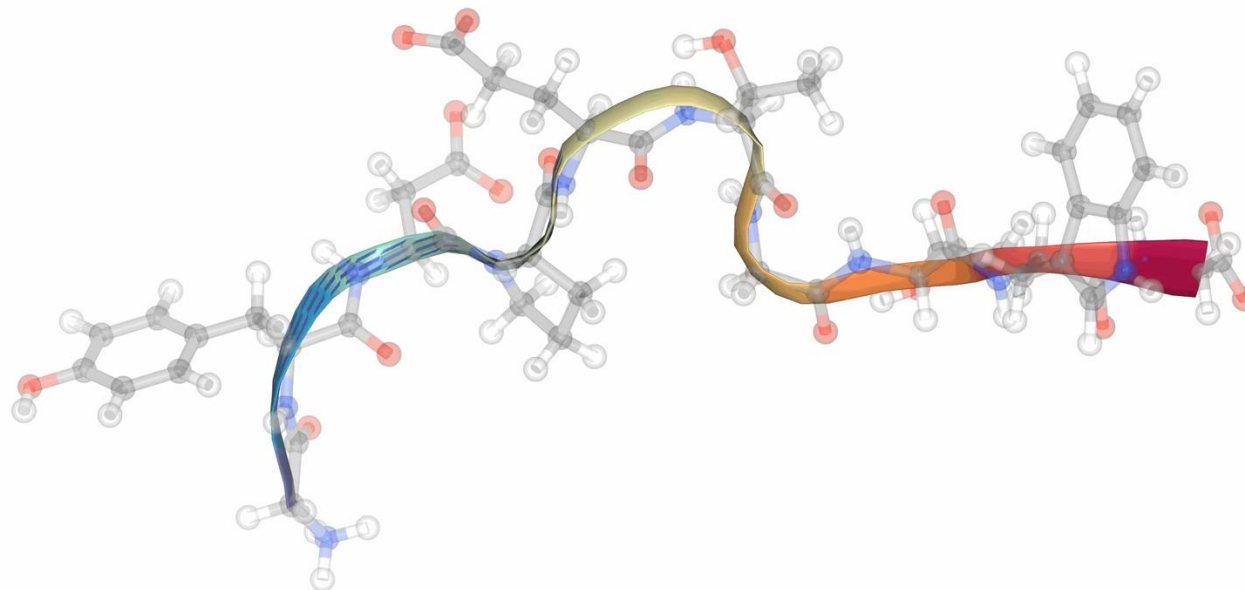
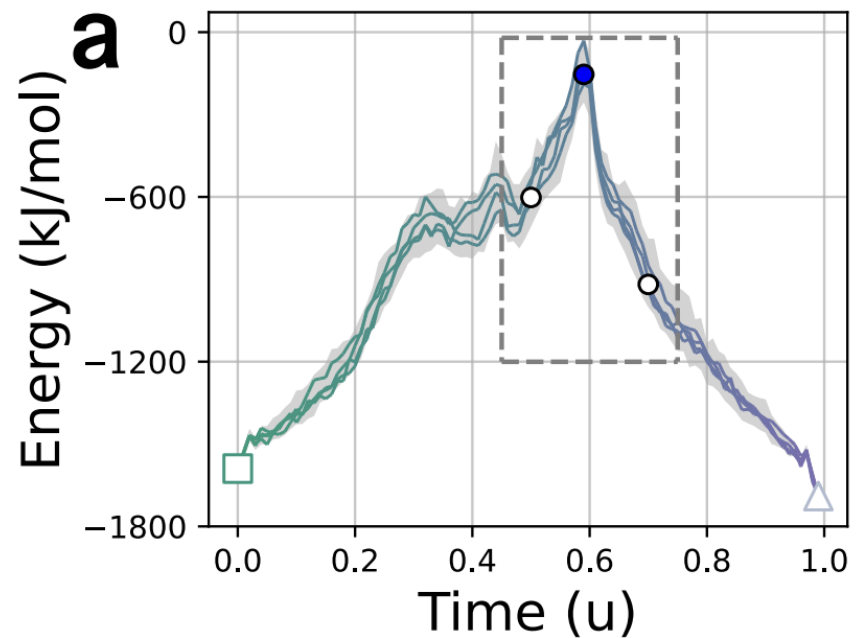
Reactive landscapes of
Nucleophilic substitution (S_N2)



Reactive landscapes of
Cycloaddition with post-TS bifurcation

Transition Path Sampling

Protein folding (Chignolin)



Adjoint-based Diffusion Sampler

Variational Inference / Stochastic Optimal Control

$$\min_u \mathbb{E} \left[\int_0^1 \frac{1}{2\sigma_t^2} \|u_t(X_t)\|^2 dt + g(X_1) \right]$$

s.t. $dX_t = [f_t(X_t) + u_t(X_t)]dt + \sigma_t dW_t, \quad X_0 \sim \rho_{\text{source}}$



Scalable Adjoint-based Method

$$u^* = \arg \min_u \mathbb{E}_{p^{\bar{u}}} [\|u_t(X_t) - \Phi_t(\mathbf{X}) \cdot \nabla g(X_1)\|^2]$$

$\bar{u} = \text{stopgrad}(u)$



Probabilistic Characteristics

$$\Rightarrow p^*(X_1|X_0) \propto p^{\text{base}}(X_1|X_0)e^{-g(X_1)}$$

[1] *Adjoint Sampling*, [ICML 2025](#).

[2] *Adjoint Schrödinger Bridge Sampler*, [NeurIPS 2025 \(oral\)](#).

[3] *Non-equilibrium Annealed Adjoint Sampler*, [NeurIPS 2025](#).

[4] *Well-Tempered Adjoint Schrödinger Bridge Sampler*, [reviewed at ICLR 2026](#).

[5] *Functional Adjoint Sampler*, [to be submitted to ICML 2026](#).

Code will be out
in next two weeks!